

PARTICULAR SPECIFICATION P5

Electrical Works

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NOTE: THIS DOCUMENT SUPERCEDES THE ORIGINAL VERSION OF THE P5 – ELECTRICAL WORKS AT THE CLOSE OF TENDER. THE CHANGES HAS BEEN HIGHLIGHTED IN YELLOW.

P5.1 OVERVIEW

- (a) This section describes the minimum requirements for the electrical works. This Specification shall be read in conjunction with all other sections of the Contract.
- (b) The electrical works (Design and Build) for Digestion Pre-treatment 1 and Sidestream Nitrogen Removal facilities areas included in the following areas:
 - (i) Sidestream Nitrogen Removal Facility (WST02);
 - (ii) Digestion Pre-treatment 1 Facility (WSD11); and
 - (iii) Interfaces between the facilities and Other contractors.
- (c) C3B1 will provide incoming power supply to the Sidestream Nitrogen Removal Facility. The Contractor shall provide the 22 kV cables (1x 3C 300mm.sq and pilot cables) from C3B1 to the Sidestream Nitrogen Removal Facility, including the termination of the 22 kV cables in the C3B1. Where required the Contractor shall also modify, replace and reconfigure the protection relay and CTs for their transformers. The Contractor shall liaise and coordinate with the Other contractors for this interface.
- (d) The Contractor shall provide the LV cables (16x 1C 630mm.sq + CPC) between WST02-TRF-0101A and WST02-TRF-0101B to WST02-LVSG-0101A and WST02-LVSG-0101B respectively, including the termination of the cables.
- (e) The C4A will provide incoming power supply to Digestion Pre-treatment 1 (WSD11) The Contractor shall provide the 400V cables (LV + CPC cables) from C4A to Digestion Pre-treatment 1 (WSD11) including the termination of cables in C4A. The Contractor shall liaise and coordinate with C4A for this interface.
- (f) The Contractor shall provide space and electrical provision in their design for future Odour Control Facility (OCF) as per Particular Specification P4 Process – Mechanical Works requirements in the Sidestream Nitrogen Removal Facility.

P5.1.2 Hazardous Area Classification

- (a) Electrical equipment and electrical installations shall comply with relevant Hazardous Area Standards when installed in areas defined in Appendix P5A – Hazardous Area Classification.
- (b) Verification of the Hazardous Area Classification shall be completed by the Contractor to ensure suitability of equipment selected by the Contractor.
- (c) The Contractor shall develop, and maintain until handover, a register of Hazardous Area Equipment.

P5.1.3 Interfacing Work with Other contractors

- (a) The attention of the Contractor is drawn to the interfacing work, liaison and coordination required with the Other contractors. The following list is by no means comprehensive and additional sequenced activities will need to be developed through co-ordination between the Other contractors together with the S.O..

(i) Civil works for the plant sitewide electrical duct banks:

- 1) The Contractor shall coordinate and interface with C3A on the electrical duct banks and provide the necessary electrical duct banks and manholes from interface point to their facilities.
- 2) The Contractor shall provide cable support system for their cables within their manholes and manholes provided by other TWRP contractors (C3A) to their facilities.
- 3) All lead-in pipes entering the buildings shall be installed with Multi Cable Transit (MCT) and all lead-in pipes in the manholes shall be sealed with cable duct sealing system.

(ii) Interfaces at C3B1

- 1) C3B1 provides power supply (at 22 kV, 3-phase, 50 Hz with N + 1 (2 x 100%) redundancy configurations) to the Sidestream Nitrogen Removal Facility. The Tender SLD shows the minimum framework structure of the electrical distribution system. The Contractor shall provide the electrical design for the Sidestream Nitrogen Removal Facility. Refer to the following sections for detailed requirements.
- 2) The Contractor shall provide the 22 kV cables from the C3B1 to the Sidestream Nitrogen Removal Facility, including the termination of 22 kV cables to the 22kV switchgear in C3B1.
- 3) The Contractor shall work, coordinate and liaise with the C3B1 contractor for the complete electrical installation including system analysis, protection coordination for the entire TWRP electrical plant. The S.O. will work with the Contractor in overseeing the interfacing works between different TWRP contractors.
- 4) The Contractor shall coordinate with the C3B1 contractor for the power turn-on including supervision of Licensed Electrical Worker (LEW) of the appropriate grade. All fee involving engaging the C3B1's LEW for the turn-on to its facility shall be included in the Contract. The S.O. will work with the Contractor in overseeing any power turn-on works between different TWRP contractors.

(iii) Interfaces with C4A contractor

- 1) The C4A provides power supply to the Digester Pre-Treatment Facility. The Contractor shall provide the electrical design for the Digester Pre-Treatment Facility. Refer to the following sections for detailed requirements.
- 2) The Contractor shall provide the LV cables from the C4A's installation to their Digestion Pre-Treatment's equipment, including the cable containment and termination of cables to the LV MCC switchgears in C4A.
- 3) The Contractor shall work, coordinate and liaise with the C4A contractor for the complete electrical installation including system analysis, protection coordination for the entire TWRP electrical plant. The S.O. will work with the Contractor in overseeing the interfacing works between different TWRP contractors.
- 4) The Contractor shall coordinate with the C4A contractor for the power turn-on including supervision of Licensed Electrical Worker (LEW) of the appropriate grade. All fee involving engaging the C4A's LEW for the turn-on to its facility shall be included in the Contract. The S.O. will work with the Contractor in overseeing any power turn-on works between different TWRP contractors.

(iv) Interfaces with C5A contractor

- 1) The Contractor shall liaise, coordinate, and provide power and control cables including cable raceways to UPS equipment provided by C5A.
- 2) The Contractor shall also provide the necessary as reference in the Particular Specification P6 ICA Works.

P5.2 SCOPE OF WORK

P5.2.1 Engagement of Licensed Electrical Worker

- (a) The Contractor shall provide services of a Licensed Electrical Worker (LEW) of the appropriate grade for the electrical works in accordance with the Specifications, Drawings and in compliance with statutory regulations. The LEW shall be responsible for all investigation, calculation, evaluation, verification, design, amendment, installation, testing, commissioning, licensing, alteration, installation, licensing, operation and maintenance, submissions and endorsement of all electrical works, drawings, calculations and test reports as and when required by the S.O., interfacing and coordination with Other contractors, statutory regulations and local authorities. These shall include coordination with Other contractors and information requested by Other contractors.

- (b) The LEW shall issue statement of safe turn-on to the S.O. for review prior to turn-on of all electrical equipment. Where interfaces with Other contractors are required, the S.O. will coordinate between the Contractor and Other contractors for all interfaces to ensure successful and safe turn-on.
- (c) Electrical drawings shall be checked and endorsed by the Contractor's LEW in accordance with the Specifications, Drawings and relevant Singapore Standards and Code of Practice.

P5.2.2 General Requirements

- (a) Overall
 - (i) The single line diagrams (SLDs) shown on the Drawings show minimum framework structure requirements with features of the electrical plant. Adequate redundancy and standby features have been indicated and these are to be retained and where necessary enhanced to achieve the reliability intended. The Contractor shall carry out a complete electrical system studies and analysis to support selection of adequately rated equipment.
 - (ii) The Contractor shall be responsible for sizing and designing all electrical equipment in accordance with local regulations, standards and codes of practice. The design shall adhere to good engineering practice and the guidelines specified in this Specification. Any costs associated with upsizing of the equipment ratings, shall be borne by the Contractor. The design shall be submitted to the S.O. for approval.
 - (iii) The Contractor shall provide appropriate motor and associated electrical equipment ratings based on the unit processes selected at no additional cost.
 - (iv) All electrical equipment provided by the Contractor shall be sized to fit within the buildings and electrical switch rooms.
 - (v) LV Switchgears directly fed from transformers shall have Automatic Transfer Control (ATC) facility as specified in the Specifications.
- (b) The submissions by the Contractor in accordance with the Specifications shall include but not be limited to the following:
 - (i) Electrical network single-line diagrams with details of circuit breakers, busbars, sizing of cables, busways, transformers, protection devices, measurement, and interlocks;
 - (ii) Single line diagrams, detailed control circuit schematic, wiring and termination diagrams for high voltage switchgear, transformers, low voltage switchgear, motor control centres, distribution boards, lighting systems, lighting distribution boards and other panels for equipment packages;

- (iii) Detailed design information for all equipment shall include relay settings to afford discrimination to the electrical network protection system between upstream and downstream equipment;
 - (iv) Electrical equipment general arrangements, panel layouts, panel sections, clearances from structures and adjacent equipment;
 - (v) Layout of all electrical components within each (front/rear) compartment;
 - (vi) Layout and design of earthing system;
 - (vii) Layout and design of lightning protection system;
 - (viii) Equipment installation, cable termination details;
 - (ix) Electrical services layouts (indoor, outdoor lighting & small power), including simulation of lighting (lux) levels;
 - (x) Cable route drawings including installation details and cross sections; and
 - (xi) Drawings detailing the necessary cable trenches, ducts, penetrations and holes and openings and fixing holes for panels, arrangement of cable trays and their supports.
 - (xii) Refer to Particular Specification P2 Preliminaries and General for additional submittal requirements
- (c) Electrical System Analysis
- (i) The Contractor shall carry out a complete analysis of the electrical distribution system for the complete Works. The Contractor shall submit these reports to the S.O. (no later than 12 months from Commencement Date), which will be provided to the Other contractors:
 - 1) One (1) report for Digestion Pre-treatment 1 Facility to be provided to C4A contractor; and
 - 2) One (1) report for Sidestream Nitrogen Removal Facility to be provided to C3B1 contractor.
 - (ii) The electrical system analysis shall be in accordance with the General Specifications. The Contractor shall note the following liaison and coordination with Other contractors and the report shall include the following but not limited to:
 - 1) The electrical system analysis shall be based on a worst-case scenario that is with the assumption that one of the pair of transformers is out of service and the bus couplers are closed at the HV/LV switchgears and downstream MCCs and MDBs.

- 2) Electrical power demand of the facility;
 - 3) Load flow study for the complete electrical installation;
 - 4) Fault level study, including motor contributions and consideration of ATC operation;
 - 5) Motor starting study (Transient Voltage Dip Study);
 - 6) Short-circuit and harmonic contribution of the facility;
 - 7) Protection coordination study with Other contractors; and
 - 8) Harmonic analysis of the system including the UPS load by C5A, and mitigation proposals to achieve compliance.
- (d) Conformal Coating of Electronic Components and Printed Circuit Boards (PCBs)
- (i) All electronic components and printed circuit boards used in all the VFD across the entire project shall be provided with conformal coating (i.e., polymeric film coating) in accordance with IEC 61086.
- (e) Switchroom Design and Equipment Layout
- (i) General
 - 1) Floor openings (including opening for future boards) for cable entry. Floor openings and equipment design shall take into consideration the location of floor beams under equipment. The Contractor shall provide and ensure that the floor openings are adequate and suitable for the final equipment installation.
 - 2) The Contractor shall provide switchgear floor frames where necessary. All unused floor openings shall be provided with aluminium chequered plates with adequate strength, suitable for man and material movement. Upon completion of cable installation and commissioning, all floor openings (used and unused) shall be sealed with a fire-stop of not less than two-hour fire rating and shall be covered with approved covers.
 - 3) All switchboards shall have a slip-proof rubber mat at least 600 mm wide in front of all MCCs and Distribution Boards. The rubber mat shall comply with IEC 61111, Class 0 for LV switchgears, MCCs and DBs and Class 3 for HV switchgears and HV motor starter panels.
 - (ii) At Digestion Pre-treatment Facility
 - 1) Electrical equipment shall be selected such that it can be accommodated in the electrical switchrooms. The Contractor shall liaise and coordinate with C4A on the equipment arrangement in the electrical switchrooms.

- 2) The Contractor shall liaise and coordinate with C4A on the floor openings for any harmonics mitigation panels in the electrical switchrooms.
- 3) The Contractor shall liaise and coordinate with C4A on the floor openings for any power factor correction panels in the electrical switchrooms for motors that required compensation.
- 4) The Contractor shall furnish the heat load calculation of their equipment selection for Other contractors to verify their air-conditioning sizing.
- 5) The Contractor shall also provide their proposed electrical equipment layout within the electrical rooms for coordination of the air-conditioning ducting routing and other equipment.

(iii) At Sidestream Nitrogen Removal Facility

- 1) All electrical switchrooms shall be air-conditioned. Refer to Particular Specification Appendix P8A Room Data Sheet for rooms that are designated as air-conditioned.
- 2) The Contractor shall furnish the heat load calculation of their equipment selection to verify their air-conditioning sizing
- 3) All electrical switchrooms shall be segregate with fire wall as specified in P8 Architectural Works.
- 4) The equipment layout to provide all round clearances of 1m for transformers and from the sides and back of switchboards, and 2m for the front of switchboards.
- 5) The Contractor shall provide gravel for all transformer yards that are oil containment pits. Provision shall be made for pumping out oil by portable pumps from the oil containment pit in case of leakage.

(f) Provision of Protective Shelters for Outdoor Electrical Equipment

- (i) The Contractor shall provide an additional protective shelter over electrical equipment to be installed outdoors. The shelter shall be inclined towards the rear and overhang the complete perimeter of the equipment by at least 200 mm. Where the shelter is larger than the stipulate size of 3m x 3m, it shall comply with the authority and code requirement.
- (ii) The covers shall be corrugated aluminium roof or corrosive resistance material shall be suitable for weather condition and coastal environment. The roofing sheet covers shall be not less than 2 mm thick with adequate bracing. The bolts and nuts shall be stainless steel sheet (grade 316) materials.
- (iii) Contact or joint of two dissimilar materials shall be provided with insulation to minimise galvanic corrosion.

- (g) Power Monitoring System
 - (i) The Contractor shall provide energy meters that shall be interfaced with the Power Monitoring System. Refer to Particular Specification P6 ICA Works and ICA Drawings for the interfaces. Refer to Drawings for locations of energy meters.
- (h) Low Voltage Distribution Transformers (22/0.415kV)
 - (i) All transformers shall be in accordance with the General Specifications and as shown on the Drawings. Prior to procurement, the Contractor shall carry out the necessary coordination works to adequately size the transformers in accordance with the final loads and equipment selection. Note that transformers shall be installed in the bays with adequate natural ventilation.
 - (ii) The transformer's radiators can be in any of the following form:
 - 1) Tubular or Flat fin radiators.
 - (iii) Design with 100% redundancy to match the configuration of the switchboards that consist of two incomers and interconnecting bus couplers.
 - (iv) The transformer kVA rating shall have at least 10% design margin on the maximum load demand without any diversity factor.
 - (v) The transformers shall be adequately K-factor rated for the harmonics. The Contractor shall calculate K-factor of the transformer in accordance with IEC 60076-20, prior to procurement of the transformers.
 - (vi) The Contractor shall provide gravel for all transformer bays with oil containment bunds of adequate capacity. Provision shall be made for pumping out using portable pumps from the oil containment pit in case of leakage. Covered transformer bays shall be verified for adequate natural ventilation.
- (i) Low Voltage Switchgear and Motor Control Centres
 - (i) All low voltage switchgear (LVSG) and motor control centres (MCC) shall be in accordance with the General Specifications and as shown on the Drawings. The Contractor shall carry out the necessary coordination to design the electrical equipment in accordance with the final loads and equipment selection. All ACBs shall have self-diagnostic features and storage of operating data.
 - (ii) All bus-couplers for LVSGs, MCCs, MDBs shall be 4P type.
 - (iii) For each size of feeder or motor starter, the Contractor shall provide one spare that is complete with all accessories. This includes VFDs that are installed within the MCC.

- (iv) The Contractor shall provide two MCCs (WSD11-MCC-0101A and WSD11-MCC-0101B, not shown in SLD) for common process equipment (not serving individual THP) including DBs for lighting and small power installation serving the THP Courtyard. The supply for the two MCCs shall be taken from WSD11-MCC-0001A (LV Switchroom 7) and WSD11-MCC-0001B (LV Switchroom 8) respectively. The configuration of these two MCCs shall complete with a coupler and M&E interlocks. The MCCs and DBs shall be located in LV Switchroom 7 and LV Switchroom 8.
- (v) The LVSGs and MCCs are categorised into the following:
 - 1) Process LVSG and MCCs:
 - These LVSGs and MCCs provide power to all equipment that are process-related equipment (i.e. indicated on the P&ID and process equipment specifications); and
 - The monitoring and controls shall interface with the SCADA through the Process Network. Refer to Particular Specification P6 ICA Works and Drawings.
 - 2) Non-process MCCs:
 - These MCCs provide power to all non-process equipment such as ventilation fans, air-conditioning, lighting and small power; and
 - These MCCs shall interface with the BMS (Building Management System) via the Non-Process Network. Refer to Particular Specification P7 Building Services and Drawings.
- (vi) All ratings shown on the Drawings are indicative minimum only. The Contractor shall provide all accessories in accordance with the Contractor's complete electrical system analysis.
- (vii) The enclosures of all switchgear and MCCs shall provide IP43 for indoor installation.
- (viii) The Contractor shall provide temperature sensing cable with fail safe features to determine high temperature or fire in all the LVSG.
- (ix) The Contractor shall provide linear heat detector system for all low voltage switchboards. For temperature sensing cable shall be inserted into the inner enclosure.
 - 1) The linear heat detector shall be capable of giving an alarm at suitable temperature. The proposed temperature shall be submitted for S.O. approval. It should be installed complete with interface module, alarm location meter and end-of-line resistor.

- 2) Each Switchboard shall have one collective indication point. The indication shall include power supply on and panel high temperature.
 - 3) The system shall send high temperature alarm signal to the SCADA. The module shall be non-latching and shall self-restore once the initiating alarms circuit clears.
- (x) Continuous On-line hot spot temperature Monitoring and Alert System for all LV Switchgears/Switchboards and MCCs:
- 1) The Contractor shall provide direct continuous hot spot temperature monitoring of the incomer, bus-coupler circuit breakers and outgoing feeders to motors and VFDs rated 75 kW and larger, large switches or breakers (rated 630A and larger). Hot spots shall be selected in proximity to the circuit breaker isolating contacts, switch contacts, VFD enclosures, busbar joints and cable termination.
 - 2) The temperature sensors shall measure temperature trends of the hot spot and upon over temperature of hot spots initiate two stage/level audio and visual alarm in the switchroom including communication with PMS through TCP/IP for similar alerts, trending and archiving. Temperature alerts shall be from rising temperature rise above ambient and difference between the 3 phase conductors. The pre-set temperature limit shall be user definable;
 - 3) The temperature sensors shall be self-powered and maintenance free measuring the direct temperature of the selected hot spots using contact measurement with wireless communication or IR (Infrared) miniature sensors. Temperature sensors shall be of accuracy not less than +/- 1% operate within the range not less than 0 to 120 degrees C. The temperature monitoring system shall be provided complete with trending storage and alarm controller units, as specified;
 - 4) Include temperature indicators, which shall have peak holding features multi-range input and 4-20 mA output signal to PMS;
 - 5) The temperature monitoring system shall have thermal monitoring algorithms with basic and advance selection. The basic function shall comprise of absolute temperature threshold for hot spot detection and phases comparison. The advance function shall have predictive threshold feature allowing for detection even at low load condition and the threshold shall adapt to the loading condition;
 - 6) The temperature monitoring system shall include software analytics and a developed temperature load map of the hot spot to provide predictive over temperature of the circuit breakers, and busbars at 50% to 100% of the rated current; and

- 7) The sensors and ancillary parts shall not reduce the electrical integrity and performance of the Switchgear and MCC during normal operations or during short circuit circuits. Electrical insulation, fault current performance, creepage and clearance distances shall not be compromised.
- (xi) Connection to Solar PV System
- 1) Feeders have been allocated in the low voltage switchgear for a solar PV system. Other contractors will provide the solar PV system and terminate cables in the low voltage switchgear.
 - 2) The Contractor shall coordinate with Other contractors for all necessary interfaces to ensure successful turn-on and connection of the solar PV system.
- (xii) Lifting Trucks for ACBs
- 1) The Contractor shall provide lifting trucks for rack-in / rack-out of the ACBs. Refer to Schedule of Prices for quantity and locations.
- (j) Low Voltage Motors
- (i) The Contractor shall provide the low voltage motors in accordance with the General Specifications. The Contractor shall select the appropriate motor to meet the load requirements. Anti-condensation heaters shall be fitted to all indoor motors rated 2.2 kW and above except hermetically sealed pump motors. Anti-condensation heaters shall be fitted to all outdoor motors.
 - (ii) The LV motors subjected to weather (rain and shine) shall be rated to IP56 and provided with rain canopy should the motor non-end drive are exposed to weather.
- (k) Low Voltage Starters and Variable Frequency Drives
- (i) All low voltage starters and variable frequency drives shall be in accordance with the General Specifications. The Contractor shall carry out the necessary coordination to select the low voltage starters and variable frequency drives for the application.
 - (ii) The low voltage starters and variable frequency drives shall be rated and sized to fit in the switchrooms.
 - (iii) Low Voltage starters shall include power factor correction facilities as required to achieve the minimum power factor stated in the General Specifications.

- (l) Local Control Stations (LCS) or Local Control Panels (LCP)
 - (i) The Contractor shall provide LCS or LCP for all motors provided under the Contract. The type of local control station to be provided shall be in accordance with the P&ID.
 - (ii) For Digestion Pre-treatment Facility, where the operation is complex and safety is critical, the provision of LCS may be omitted and subject to S.O.'s approval.
 - (iii) For process equipment, the indications, switches and pushbuttons on the LCS or LCP shall be as shown on the P&ID. Local/Remote selector switch shall be key-operated switch.
 - (iv) The emergency push button shall be located externally of the enclosure suitable for weather condition and corrosive environment.
 - (v) All LCS or LCP enclosure shall be made of SS316 stainless steel material and where it is located outdoor it shall be IP55 rated with rain canopy.
- (m) Main Distribution Boards (MDBs) and Distribution Boards (DBs)
 - (i) All MDBs or DBs where located outdoor shall be made of stainless steel SS316 enclosure material and rated to IP55 with rain canopy.
 - (ii) The Contractor shall design and provide the necessary downstream distribution boards that shall provide final outgoing circuits to the equipment. This shall include providing adequate ratings to fulfil the power distribution requirements to final equipment.
 - (iii) The Contractor shall design and provide the downstream distribution boards that are located in the field for final branch circuits to all equipment.
 - (iv) The Contractor shall provide 20% spare MCBs for all Main Distribution Boards and distribution boards.
 - (v) All MDBs shall be equipped with mechanical interlocks to prevent parallel supply connection.
- (n) Cables
 - (i) The Contractor shall size and provide all cables in accordance with the complete electrical system analysis and relevant local and international codes. This shall be submitted to the S.O. for approval.
 - (ii) All power and control cables shall be armoured cables for mechanical protection. The cables shall be in accordance with the General Specification. The Contractor shall refer to the Drawings on the type of power cables to be used for the respective application.

(o) Cable Raceways and Ductbanks

- (i) The Contractor shall provide the cable ladders, trays, raceways as required to provide power supply to all equipment.
- (ii) To size and provide the cable raceways space for the future THP panel's main incoming source cables from LV Switchroom 5 to LV Switchroom 7 and from LV Switchroom 6 to LV Switchroom 8.
- (iii) The Contractor shall provide the cable routing, raceways and support systems to all equipment. The Contractor shall study the general arrangement of the propose equipment and its cable termination box location and the routing and the drop off point accordingly to suit the space provided. The Contractor shall take note of the required bending radius of the cables and all adjustment of the cable route and the equipment layout. Any low-lying containments shall be equipped with covers protection and steps for crossing over and alternatively to provide heavy duty uPVC embedment pipes between pipe rack or cable support to the equipment. All cost for a complete installation shall be deemed to be included in the Contract cost.
- (iv) All outdoor/exposed cable trays or cable ladders subjected to weather conditions shall be provided with cable trays covers. All cable trays, cable raceways and covers are subject to S.O. approval.
- (v) Separate raceways shall be used for different voltages. i.e. low voltage and extra low voltage.
- (vi) All raceway materials shall be stainless steel SS316 including cable ladders, trays, trunking, conduits, framing channels and supports.
- (vii) Outdoor cable ladders, trays and raceways: The cable ladders, trays and raceways shall be insulating (non-metallic) made of homogeneous thermoplastic raw material, continuous or perforated, for support, protection and conduction of cables for outdoor installations. Cable ladder and tray tested according to IEC 61537, with Impact resistance: 20J according to IEC 61537, Protection against mechanical damage with cover fixing piece. Minimum dielectric strength 14kV/mm according to IEC 60243-1. Suitable for outdoor use with UV exposure Weathering test and "Resistant to corrosion and with exposure to the UV". "Non-flame propagator" according to IEC 61537 and with LOI Oxygen Index according to ISO 4589:1996 $\geq 45\%$, and Glow-wire test at 960°C according to IEC 61537. All hanging rods, fasteners and accessories to be at least SS316. Fastener of the same thermoplastic material and of the same strength (or better grade) to the stainless steel fasteners can be considered subject to S.O. approval.
- (viii) The Contractor's Professional Engineer (PE) shall endorse the cable support systems, floor and wall openings and verify the structural integrity of the system.

- (ix) All building penetrations including floor slab, ceiling and roof shall have fire-stop and sealing in accordance with the Authorities' requirements.
 - (x) Where lead-in pipes for building penetrations to the exterior of the building are required, the Contractor shall provide multi-cable transits (MCTs) with water-proofing to prevent ingress of groundwater.
 - (xi) All pipe sleeves including used as well as spare sleeves shall be sealed using sealants to prevent water ingress through the pipe sleeves and to avoid flooding of the manholes.
- (p) Earthing System
- (i) The Contractor shall provide an earthing system in accordance with the General Specifications and Singapore Standard SS 551.
 - (ii) The Contractor shall provide a separate electrical and instrument (XLV) earth with a surge diverter that shall provide equipotential bonding under transient conditions.
 - (iii) The Contractor shall provide the necessary cable containment for the earthing cables and copper tape.
 - (iv) The Contractor shall submit drawings and calculations with LEW endorsement and submit to S.O. for approval before installation.
 - (v) The Contractor shall provide an earthing system for Sidestream Nitrogen Removal and Digestion Pre-treatment 1 facilities.
- (q) Lightning Protection System
- (i) The Contractor shall provide the lightning protection system in accordance with the General Specifications and Singapore Standard SS 555 and submit to S.O. for approval. Installations with sensitive electronic equipment shall have lightning earth resistance less than 1 ohm and then bonded to the electrical earthing system.
 - (ii) The Contractor shall submit the lightning protection design drawings and calculations with LEW endorsement for all structures to S.O. for approval before installation.
 - (iii) The Contractor shall provide the lightning protection system for Sidestream Nitrogen Removal and Digestion Pre-treatment 1 facilities.
 - (iv) The Contractor shall liaise, coordinate, protect and made good to any lightning installation that has already been installed by C4A in the course of their equipment installation.
 - (v) Lightning warning system will be provided by Other contractors.

(r) Lighting

- (i) The Contractor shall provide lighting at THP Courtyard for Digestion Pre-treatment 1. The lighting within Biosolids Building will be provided by Others.
- (ii) The Contractor shall provide lighting for the entire Sidestream Nitrogen Removal Facility.
- (iii) The Contractor shall use efficient lamps, luminaires, including lighting control, and energy saving measures in accordance with the General Specifications and SS 530. The minimum lux levels are listed in Table P5.2-2-1.
- (iv) The Contractor shall provide the lighting design complying with relevant local codes and required lux levels. The Contractor shall submit the illumination calculations to the S.O. for approval. The measurement of lux levels at site shall be carried out by the Contractor in the presence of the S.O. The lighting fixtures installed shall take into account the hazardous area classification and the appropriate lighting fixtures shall be provided to suit the requirements.
- (v) The lighting requirements comprises of the general lighting, emergency and maintenance lighting. The areas that require maintenance lighting are specified in the subsequent sections. Lighting fittings shall be mounted on walls furthest from external facing railings of staircases and at a height that do not require the use of ladders. The use of high bay lightings shall be reduced where possible and consider the use of pole mounted light fittings on walkways.

Table P5.2-2-1: Required Lux Levels

Type of Location	Area	Lux levels (Average)
Indoor	General walkway areas	100
Indoor	General staircase areas	150
Indoor	Store rooms	100
Indoor	Bathrooms, washrooms and toilets	200
Indoor	Electrical and ICA Equipment Rooms	200
Indoor	Operating areas (Plant Areas)	200
Indoor	Control rooms / Operations room	500
Indoor	Offices and meeting rooms	500

Type of Location	Area	Lux levels (Average)
Indoor	Laboratories	500
Outdoor	General walkway areas (Exclusive for pedestrians)	30
Outdoor	Roads, streets, loading and unloading bays	30
Outdoor	Operating areas	200

(vi) A single lighting circuit shall not have more than ten (10) light fixtures connected. Separate circuits shall be provided for emergency and exit lights. Each lighting circuit shall have individual residual current devices.

(vii) Selection of Lighting Fixtures

- 1) LED lighting shall be used in all areas;
- 2) The appropriate lighting fixtures shall be selected in accordance with the Hazardous Area Classification;
- 3) All light fittings shall have appropriate protection rating for the intended location;
- 4) Task lighting flood lights shall be provided adjacent to equipment such as pumps and motors;
- 5) Flood lights shall be provided on electric overhead travelling cranes; and
- 6) Green coloured lights shall be provided at the safety showers.

(viii) Lighting control circuits shall minimise the energy usage in accordance with SS 530. Circuit/switching or dimming shall be provided so that:

- 1) Lighting in task areas larger than 10m² can be reduced when the task is not performed or when the task is relocated;
- 2) Lighting in task areas can be controlled automatically by having only minimum lux level when the task is not performed;
- 3) Lighting can be reduced when a space is empty and not being used;
- 4) Lighting can be turned off (manually and automatically) where daylighting is adequate and can be used without discomfort;

- 5) External lighting shall be controlled by time switches, photo-cells or other devices to turn it off during the hours such lighting is not necessary; and
 - 6) Contractor shall submit lighting schedule and lighting schematic diagram by including multiple switches for the process areas based on the area / zone, prior to installation.
- (ix) Lighting control circuits shall minimise the energy usage in accordance with SS 530. Circuit/switching or dimming shall be provided so that:
- 1) External lighting for the outdoor operating areas shall be controlled by time switches, photo-cells or other devices to turn it off during the hours such lighting is not necessary; and
 - 2) Contractor shall submit lighting schedule and lighting schematic diagram by including multiple switches for the process areas based on the area / zone, prior to installation.
- (x) The Contractor shall refer to Table P5.2-2-2 for the minimum requirements that the Contractor shall provide for general lighting and maintenance lighting. This shall be read in conjunction with the General Specification G5.22 Lighting System.

Table P5.2-2-2: Criteria for Selection and Installation of Lighting Fixtures for TWRP

Type of Room / Area	General Lighting	Maintenance Lighting	Remarks
Battery Rooms	F4 & F5	-	-
Cable Chambers	F4 & F5	-	-
Control Room / Operations Room	F4 & F5	-	-
Corridor	F4 & F5	-	-
Crane Control Room	F4 & F5	-	-
Electrical Riser	F4 & F5	-	-

Type of Room / Area	General Lighting	Maintenance Lighting	Remarks
Electrical Switchrooms	F4 & F5	-	-
Fire-fighting Lobby	F4 & F5	-	-
Fire Command Centre	F4 & F5	-	-
Fire Water Pump Room	F4 & F5	H5	Floodlights shall be provided for maintenance lighting.
Gas-cylinder Room	F4 & F5	-	-
Maintenance Floor	F4 & F5	H5	Floodlights shall be provided for maintenance lighting.
Meeting Room	F4 & F5	-	-
Non-Process XLV Room	F4 & F5	-	-
Outdoor Process Areas (e.g. Digesters, Odour Control Facility)	F4 & F5	H5	Floodlights shall be provided for maintenance lighting.
Pantry	F4 & F5	-	-
Process XLV Room	F4 & F5	-	-
Process Server Room	F4 & F5	-	-
Smoke-stop Lobby	F4 & F5	-	-
Staircase	F4 & F5	-	-

Type of Room / Area	General Lighting	Maintenance Lighting	Remarks
Standby Generator Room	F4 & F5	H5	Floodlights shall be provided for maintenance lighting.
Storage Room	F4 & F5	-	-
Transformer Bay	F4 & F5	H5	Floodlights shall be provided for maintenance lighting.
Toilets	F4 & F5	-	-
UPS Room	F4 & F5	-	-

(s) Switch-socket Outlets (SSO)

- (i) Socket outlets (with appropriate weatherproof enclosure and padlocks) shall be installed for providing small power to portable equipment.
- (ii) Each circuit of SSO shall be provided with individual residual current devices.
- (iii) The Contractor shall provide the required SSOs to feed power supply to all equipment provided. In addition, the Contractor shall provide SSOs at the locations in accordance with the following guidelines:
 - 1) In operations and control rooms: at least two (2) twin 13 A socket outlet for each workstation and as required for all equipment (e.g. printers etc.), in addition, the Contractor shall provide one (2) twin 13 A socket outlet for every 5 m;
 - 2) In general walkway areas and corridors: at least one (1) twin 13 A socket outlet for every 15 m;
 - 3) In process areas (indoor and outdoor areas): at least one (1) twin 13 A socket outlet (weatherproof) for every 15 m and one (1) 3-phase 30 A socket outlet (weatherproof) for every 30 m;
 - 4) In external areas (perimeter of buildings): at least one (1) twin 13 A socket outlet (weatherproof) for every 15 m;

- 5) In all rooms other than operations and control rooms: at least one (1) twin 13 A socket outlet with a maximum spacing of 3 m; and
 - 6) And all socket outlets (1-phase or 3-phase) required at the workstations, desks and control room workstations.
- (iv) Lighting Distribution Boards
- 1) The Contractor shall provide all necessary lighting distribution boards to feed power supply to all lighting circuits. Segregate lighting DBs between floors.

P5.2.3 BCA Green Mark

- (a) The Contractor shall provide energy efficient lighting to achieve an overall of at least 40% improvement in Lighting Power Density (LPD) as specified in SS 530. The Contractor shall ensure that the lux levels and lighting controls shall be designed in accordance with SS 531.
- (b) The Contractor shall provide the following documents and detailed calculations to submit to BCA for Green Mark Certification for the buildings under this Contract:
 - (i) Lighting layout plans for interior, exterior, Façade, Landscape area etc, as required;
 - (ii) Lighting schedules showing the number of lighting fixtures, types and location;
 - (iii) Calculation of proposed lighting density (LPD) and percentage energy savings;
 - (iv) Tabulation showing designed lux level and minimum code requirement;
 - (v) Product information of Luminaries (Light fittings and Ballast);
 - (vi) The Contractor shall provide high frequency ballast for all applicable lightings;
 - (vii) All product details and technical information of HF (High Frequency) electronics ballast and list of lightings which are used with HF electronic ballast;
 - (viii) Lighting lux level simulation for all areas;
 - (ix) Technical data for the luminaires used including driver specification and ballast details;
 - (x) Testing and commissioning report of lighting system including automatic control and lux levels achieved;
 - (xi) Clear copy of PO/DO of all luminaires used; and
 - (xii) Tabulation of the % efficiency of all distribution transformers, and all motors installed under this Contract.

P5.2.4 Manufacturers and Products

- (a) Where two or more units of the same class of material or equipment are required, the Contractor shall provide products of the same manufacturer. Component parts of materials or equipment need not be products of the same manufacturer.

P5.2.5 Works at the Sidestream Nitrogen Removal and Digestion Pre-treatment 1 Facilities

- (a) Low Voltage Distribution Transformers (22/0.415 kV)
- (i) The Contractor shall provide the transformers listed in Table P5.2-2-3 and as shown on the Drawings.

Table P5.2-2-3: LV Distribution Transformers for Sidestream Nitrogen Removal

Equipment Tag	Location
WST02-TRF-0101A	Sidestream Nitrogen Removal Building Transformer Bay 1
WST02-TRF-0101B	Sidestream Nitrogen Removal Building Transformer Bay 2

- (b) Low Voltage Switchgear and MCC

- (i) The Contractor shall provide, as the minimum, the low voltage switchgear and MCC listed in Table P5.2-2-4 and Table P5.2-2-5.

Table P5.2-2-4: LVSG for the Sidestream Nitrogen Removal Facility

Equipment Tag	Location
WST02-LVSG-0101A	Sidestream Nitrogen Removal building LV Switchroom 1
WST02-LVSG-0101B	Sidestream Nitrogen Removal building LV Switchroom 2

Table P5.2-2-5 MCCs for Sidestream Nitrogen Removal and Digestion Pre-treatment Facilities

Equipment Tag	Location	Description
WST02-MCC-0101A	Sidestream Nitrogen Removal Building LV Switchroom 1	Process MCC
WST02-MCC-0101B	Sidestream Nitrogen Removal Building LV Switchroom 2	Process MCC

Equipment Tag	Location	Description
WST02-MCC-0102A	Sidestream Nitrogen Removal Building LV Switchroom 1	Process MCC
WST02-MCC-0102B	Sidestream Nitrogen Removal Building LV Switchroom 2	Process MCC
WST02-MCC-1001A	Sidestream Nitrogen Removal Building LV Switchroom 1	Non-Process MCC
WST02-MCC-1001B	Sidestream Nitrogen Removal Building LV Switchroom 2	Non-Process MCC
WSD11-MCC-0001A	Biosolids Building LV Switchroom 7	Process MCC
WSD11-MCC-0001B	Biosolids Building LV Switchroom 8	Process MCC
WSD11-MCC-0101A	Biosolids Building LV Switchroom 7	MCC (common process equipment, lighting and small power)
WSD11-MCC-0101B	Biosolids Building LV Switchroom 8	MCC (common process equipment, lighting and small power)

(c) Main Distribution Boards

- (i) The Contractor shall provide the main distribution boards, as a minimum, as shown in Table P5.2-2-6. The Contractor shall complete the design from the Main Distribution Board to the final loads and equipment, including providing downstream distribution boards and sub-distribution boards for Sidestream Nitrogen Removal and as required for Digestion Pre-treatment Facility (not mentioned below).

Table P5.2-2-6: MDBs at Sidestream Nitrogen Removal Facilities

Equipment Tag	Location	Description
WST02-MDB-0001A	Sidestream Nitrogen Removal building LV Switchroom 1	Process
WST02-MDB-0001B	Sidestream Nitrogen Removal building LV Switchroom 2	Process
WST02-MDB-1001A	Sidestream Nitrogen Removal building LV Switchroom 1	Non-Process
WST02-MDB-1001B	Sidestream Nitrogen Removal building LV Switchroom 2	Non-Process

(d) Cables

- (i) The Contractor shall provide all cables in his scope of works.

(e) Cable Routing and Raceways

(i) Incoming 22 kV Cables to the Sidestream Nitrogen Removal Facility

- 1) The Contractor shall provide the incoming 22 kV cables to the Sidestream Nitrogen Removal Plant and the cable shall be installed in the duct banks and manholes that are provided by Other contractors.
- 2) The Contractor shall provide the duct banks from the contract boundary of C3A to the 22/0.415 kV transformers in the Sidestream Nitrogen Removal Facility. The Contractor shall coordinate with C3A contractor for the installation of his 22 kV cables.

P5.2.6 Testing and Commissioning

- (a) The Contractor shall comply to Particular Specification P10 Plant Performance Testing and General Specification G2.26 Testing and Commissioning.

(b) Factory Acceptance Test

- (i) All the routine tests carried out at the Factory Acceptance Test shall be repeated at the site commissioning tests including circuits directly or indirectly connected to the switchboard. This shall include all control and power cables, interconnecting busbar, earthing systems, functional test of the switchgears and motor starter linked to the outgoing circuits and final sub-circuits for compliance with the Specifications.
- (ii) Testing of all equipment at the manufacturers' works including the supply of all testing instruments and arrangements for witnessing by S.O. and the Board.

(c) Site Acceptance Test

- (i) Installation, testing and commissioning of all electrical equipment, including the supply of all instruments, cabling works, etc.;
- (ii) Every switchboard at each process centre shall be tested individually with completion of forms C (Safe to Turn-On) and D (responsible for O&M) similar compliance with SP requirements, before turn-on of electrical supply. The forms shall be signed off by the Contractor's LEW in the appropriate voltage category supervising the testing and commissioning. All testing results and forms shall be submitted to S.O. before turn-on;
- (iii) Proposed protection setting design for the electrical equipment and co-ordination with existing upstream system, settings endorsed by electrical PE/LEW;

- (iv) Provide live thermal scans for all the HV (if applicable) and LV switchboards and terminations during site commissioning test; and
- (v) Testing of electrical equipment shall comply with General Specifications, manufacturer's recommendations, statutory requirements and as required by the LEW, before testing, commissioning and turn-on. All test results shall be submitted to the S.O. before turn-on. The Test Reports shall be certified by the LEW.
- (d) The Contractor shall coordinate with Other contractors for plant testing and commissioning and shall comply with General Specification G2.26 Testing and Commissioning.

P5.2.7 As-Built Drawings

- (a) All drawings and documentation shall comply with General Specifications and all submissions shall be endorsed by LEW.
- (b) Single line diagram shall be endorsed by LEW and framed in each electrical room for all switchgears, distribution boards and motor control centres.

P5.2.8 Operations and Maintenance (O&M) Information

- (a) The O&M information shall include the following information as a minimum for each electrical item and system:
 - (i) Detailed report of Electrical System Analysis including load flow study, short-circuit study and protection coordination shall be submitted to S.O. as part of O&M Manuals;
 - (ii) Cable sizing calculations of all cables shall be submitted to S.O. as part of O&M Manuals;
 - (iii) Description of operation;
 - (iv) Equipment setting-up and dismantling details;
 - (v) Complete Specifications;
 - (vi) List of manufacturers, model numbers and ratings of components;
 - (vii) Range of settings and recommended settings;
 - (viii) Test certificates for components;
 - (ix) Sample commissioning test results;
 - (x) List of recommended spares;
 - (xi) Set of Schematic, wiring and layout diagrams; and

- (xii) As-constructed drawings.
- (b) Refer to Particular Specification P2 Preliminaries and General for additional submittal requirements.
- (c) The Contractor shall prepare Operations and Maintenance information for all electrical equipment and submit in the prescribed format. The O&M information shall be in accordance with General Specifications.
- (d) The information shall be revised incorporating the S.O.'s recommendations/comments and commissioning test results. The revised manuals shall be resubmitted as final as per the Specifications.

P5.3 APPENDIX

- (a) APPENDIX P5A – TWRP – Hazardous Area Classification

END OF SECTION